

Subject Code & Name: 25AF10THOEM05D - FLUID MECHANICS

- 11 Dimensional homogeneity means:
 a) Different units b) Dimensions on both sides of the equation are same
 c) Units do not matter d) Only mass dimension matters
- 12 Buckingham π -theorem gives:
 a) Number of experiments b) Dimensionless groups
 c) Only length dimensions d) Error calculations

Q2. Solve the following.

- A) The space between two parallel plates 5 mm apart is filled with crude oil. A force of 20 N is required to drag the upper plate at a constant velocity of 0.8 m/s. The lower plate is stationary. The area of the upper plate is 0.09 m². Determine:
 (i) The dynamic viscosity and (ii) The kinematic viscosity of the oil in stokes if the specific gravity of oil is 0.9
- B) State and derive Pascal's law. 6

Q3. Solve the following.

- A) Explain the following flows with example (any two) 6
 i) Steady and unsteady flow, ii) Uniform and non uniform flow,
 iii) Laminar and Turbulent Flow.
- B) Explain path line, stream line and streak line with neat sketch and state the condition under which all three coincide. 6

Q4. Solve Any Two of the following.

- A) An orifice meter of 300 mm pipe diameter and 150 mm orifice diameter is fitted in a pipeline of carrying oil of relative density 0.9. The differential manometer connected across the orifice shows a reading of 500 mm of mercury. Assuming the C_d to be 0.64, determine the rate of flow of oil through the pipe.
- B) Explain Pitot tube. How is it used to measure velocity of flow at any point in a pipe using Bernoulli's equation? 6
- C) Find an expression for measuring discharge of fluid passing through a pipe with venturimeter. 6

Q5. Solve Any Two of the following.

- A) A flat plate 2m $\times$ 1m moves at 30 kmph in a stationary air of density 1.15 kg/m³. If Coefficients of drag and lift are 0.15 and 0.75 respectively, Determine drag force, lift force and resultant force on the plate and orientation of resultant force.
- B) Explain the following terms. 6
 i) Angle of Attack ii) Streamline and Bluff Body iii) Magnus Effect
- C) Explain the concept of separation of boundary layer with a neat sketch. 6

Q6. Solve Any Two of the following.

- A) What is dimensional analysis? Explain the term dimensional homogeneity. 6
- B) State Buckingham's π -theorem. Describe the step-by-step procedure used to obtain dimensionless π -terms. 6
- C) Explain any three commonly used dimensionless numbers with their physical significances. 6

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